



PHYSICAL EDUCATION STUDIES

Stage 2

WACE Examination 2012

Marking Key

Marking keys are an explicit statement about what the examiner expects of candidates when they respond to a question. They are essential to fair assessment because their proper construction underpins reliability and validity.

When examiners design an examination, they develop provisional marking keys that can be reviewed at a marking key ratification meeting and modified as necessary in the light of candidate responses.

Section One: Multiple-choice

14% (20 Marks)

Question	Answer
1	b
2	c
3	c
4	a
5	c
6	c
7	a
8	b
9	d
10	d
11	b
12	b
13	a
14	b
15	b
16	d
17	a
18	c
19	c
20	a

Section Two: Short answer

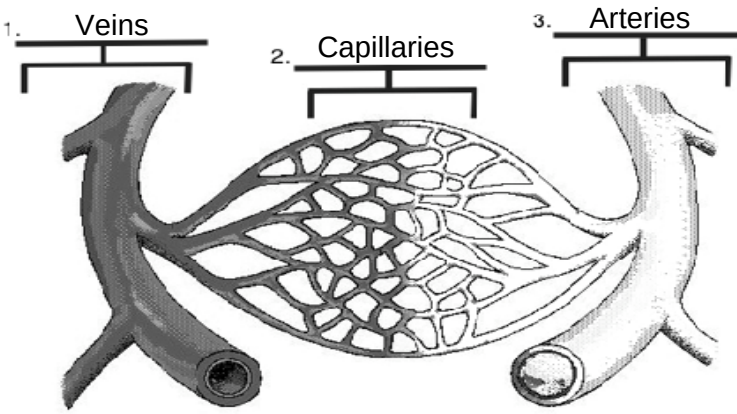
35% (50 Marks)

Question 21

(5marks)

The following diagram represents the location where gas is exchanged in the circulatory system. The grey side represents de-oxygenated blood and the white side represents oxygenated blood.

- (a) Insert labels for the **three** structures indicated in the above diagram. (3 marks)

Description	Marks
	1-3
	Total 3

- (b) Describe how gas exchange occurs at the alveoli. (2 marks)

Description	Marks
<u>Diffusion</u> of oxygen and carbon dioxide <i>or</i> appropriate description of oxygen and carbon dioxide movement across the membrane	1
Due to <u>partial pressure</u> difference (or from <u>high to low</u> concentration)	1
	Total 2

Question 22

(6 marks)

Skeletal muscle tissue has four major characteristics. One of these characteristics is excitability. Excitability describes muscle tissue's ability to respond to a stimulus. Name and describe the other **three** characteristics of muscle tissue.

Description	Marks
Elasticity:	1
return to original length/shape after extension	1
Contractibility:	1
ability to contract	1
Extendibility/extensibility:	1
stretch beyond resting length	1
	Total 6

Question 23

(6 marks)

Using the images below:

- (a) Identify with an 'X' where each activity fits on the associated continuum of skill classification. (3 marks)

Description	Marks
OPEN ←(-----X-----)-----→ CLOSED	1
DISCRETE ←-----(------X-----)→ CONTINUOUS	1
FINE ←-----(------X-----)→ GROSS	1
*Anywhere within an appropriate range is acceptable	
	Total 3

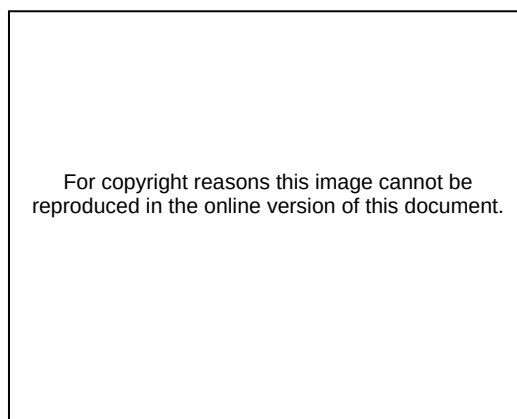
- (b) Justify the reason for the position on the continuum of skill classification you chose for each skill. (3 marks)

Description	Marks
The batter must <u>react</u> to ball; must continually <u>adapt</u>	1
The runner has not distinct start and finish; repeated gait cycle	1
The pitcher is using the entire body; large muscle groups	1
	Total 3

Question 24

(6 marks)

Below is a picture of Sally Pearson who competes for Australia in the 100 metre hurdles. Her personal best time is 12.28 seconds. She is exceptionally skilful in not letting her trail leg hit the hurdle as she passes over them. Using the image, complete the following table in relation to the trail leg as pictured.



Description			Marks
Joint	Articulating bones	Movement	
Hip	1. Femur 2. Pelvis	Abduction or flexion or circumduction	1–3
Knee	1. Tibia 2. Femur	Flexion	1–3
			Total 6

Question 25

(7 marks)

Caitlin Basset is a member of the West Coast Fever Netball team. As an athlete, she is in the Autonomous Stage of the Fitts and Posner model of motor learning. During a game of netball, she is fluent in her execution of skills.

- (a) Describe **three** other characteristics of this stage. (3 marks)

Description	Marks
Any 3 of: Responds automatically with skills appropriate for the situation Makes infrequent skill errors Has spare attention to devote to other tasks Speed and efficiency of skill is increased Uses feedback precisely Relies more on internal feedback Benefits from self analysis Focus is more tactical/strategic rather than skill focused Skills can be performed without conscious thought	1–3
	Total 3

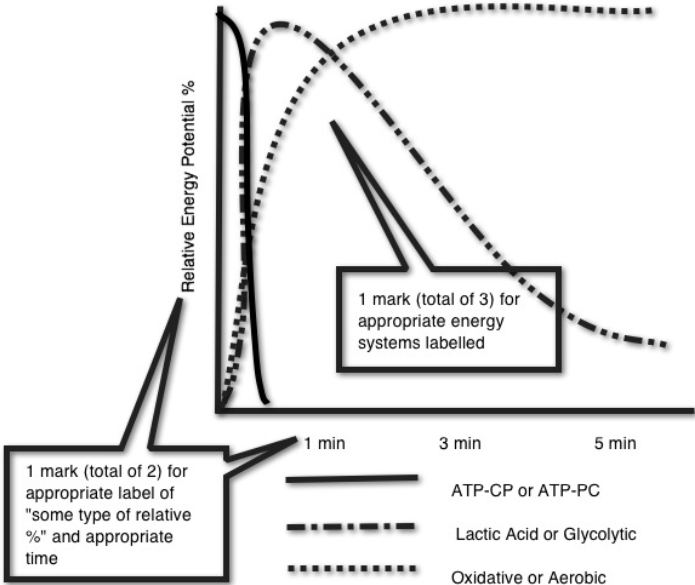
- (b) Muscular endurance and lower body muscular power are important fitness components for Caitlin. Define these components and name a type of training that would improve each fitness component. (4 marks)

Description	Marks
Muscular Endurance – ability to repeatedly contract a muscle with force	1
Associated training type, any of: resistance training; interval; continuous; circuit; fartlek	1
Muscular power – rapid application of muscular force in a short <u>time</u>	1
Associated training type, any of: resistance training; plyometrics	1
	Total 4

Question 26

(5 marks)

In the space below, draw a graph (labelling each axis) that represents the potential relative energy contribution and timing of the three energy systems for a running event that lasts approximately 5 minutes.

Description	Marks
 1 mark (total of 2) for appropriate label of "some type of relative %" and appropriate time 1 mark (total of 3) for appropriate energy systems labelled Legend: — ATP-CP or ATP-PC - - - Lactic Acid or Glycolytic ... Oxidative or Aerobic *Please allow for the following variation for answers/lines in the graph • ATP-CP (line peak 5–10 seconds and not exceed 30 seconds) • Lactic Acid (peak between 30–60 seconds) • Oxidative cross before 2 minutes	1–5
	Total 5

Question 27

(5 marks)

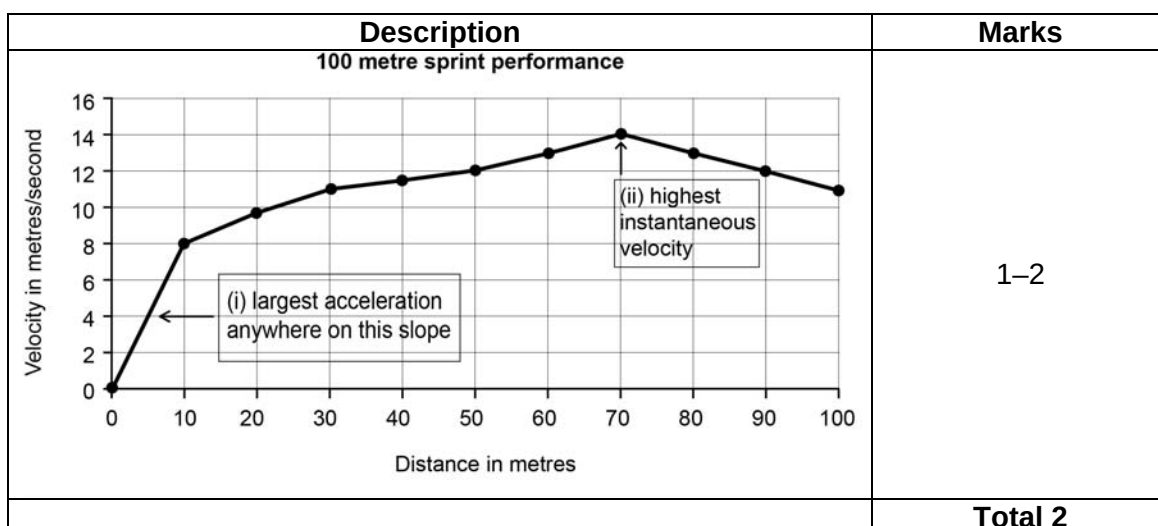
In an Olympic year, the 100 metre sprint is of great interest to many people, as it 'crowns the fastest human'. Use the diagram below to help you understand how one of these runners completes the race.

- (a) Is the average velocity or peak instantaneous velocity higher for a 100 metre runner?
(1 mark)

Description	Marks
Peak instantaneous	1
	Total 1

- (b) Label on the graph where you would find:

- (1) the largest acceleration and
(2) the highest instantaneous velocity.
(2 marks)



- (c) Justify the location you selected for (1), the largest acceleration and (2), the highest instantaneous velocity.
(2 marks)

Description	Marks
1. The greatest change in <u>velocity</u> or fastest increase in speed or gradient of the graph is the steepest	1
2. highest point in velocity graph or highest speed in split time	1
	Total 2

Question 28

(4 marks)

Nideffer's model identifies four main attentional styles. Write the attentional style required for each of the following situations.

Description		Marks
Style	Situation	
Broad-Internal	Bowler deciding what type of spin to place on his next delivery	1
Narrow-External	A tennis player focusing on returning a serve	1
Narrow-Internal	A golfer mentally rehearsing his drive from the tee	1
Broad-External	Attacker dribbling ball down field checking positions of opposition players	1
		Total 4

Question 29

(6 marks)

On the diagram below, name and label the following muscles:

- (a) The muscle known as the most powerful hip extensor. (2 marks)
- (b) The muscle responsible for dorsiflexion. (2 marks)
- (c) The muscles responsible for elbow flexion. (2 marks)

Description	Marks
For copyright reasons this image cannot be reproduced in the online version of this document.	1–6
*may label either left or right side	
*name or label only (1 mark each)	
	Total 6

Section Three: Extended answer

21% (30 Marks)

Question 30

(15 marks)

Aaron is competing in the opening round of the shot put competition at the Olympic Games. His first attempt resulted in a performance well below his usual standard. After his first attempt, Aaron said to himself that he “felt as if he didn’t put enough force into the throw” as he saw the distance up on the scoreboard, and at the same time, his coach told him he needed to “release the shot in an extended position with a higher right shoulder”. The images below represent the path of the shot put during his first and second attempts.

- (a) Name and describe the **three** components of projectile motion that influenced his results. (6 marks)

Description	Marks
Angle of release	1
Description of different angles of release during two attempts	1
Velocity of release or velocity of projectile	1
The velocity or speed at which he threw the shotput during the two attempts or force related to velocity	1
Height of release	1
Description of him having a higher or different release point	1
	Total 6

- (b) Name and describe **three** types of feedback that were provided by the coach and Aaron during his performance. (6 marks)

Description	Marks
Description related to how Aaron ‘felt’; ‘proprioception’; ‘force applied’	1
Intrinsic (internal)	1
Description related to ‘scoreboard’; ‘distance thrown’	1
Terminal or *extrinsic (external) or nonverbal	1
Description related to ‘coaches verbal feedback of ‘extended position...’	1
Verbal or *extrinsic (external) or terminal	1
*Can receive 1 mark for extrinsic only once	
	Total 6

- (c) Using specific examples from the scenario on page 18, justify how each type of feedback received by Aaron related to the changes in performance that occurred between his first and second attempts. (3 marks)

Description	Marks
Intrinsic (or description of feedback) – related to him ‘ <u>applying more force</u> ’ to <u>increase projectile velocity</u>	1
Terminal or extrinsic (or description of feedback) – related to knowing his <u>result to improve upon</u>	1
Verbal or extrinsic (or description of feedback) – related to him <u>changing his release height and release angle</u>	1
	Total 3

Question 31

(15 marks)

In 2011, Australian Craig Alexander won his third Hawaiian Ironman in the record time of 8 hours 3 minutes and 56 seconds. This race requires athletes to swim 3.8 km, cycle 180 km and then run a marathon (42 km).

- (a) Identify and describe **five** principles of training for continued success in the Hawaiian Ironman event. (10 marks)

Description	Marks
Any 5 principles, one mark for principle, 1 mark for description	1–10
Progressive overload	
to gradually increase volume or training load (frequency and duration)	
Reversibility	
Decrease in training causes loss of training adaptation or 'use it or lose it'	
Specificity	
his training is specific to his event of swimming, cycling and running	
Frequency	
the number of sessions he has per week or per day	
Intensity	
the relative difficulty or percentage load of training (e.g. %HR or %RM)	
Duration	
length of time of each session or training cycles	
	Total 10

- (b) Part of Craig's race plan is to maintain energy stores to reduce the onset of fatigue. If Craig fails to follow his supplementation plan he will be unsuccessful in his race. Explain, from the onset of his race, how his body will utilise each energy source through the race and the physiological consequence of only drinking water during the race.(5 marks)

Description	Marks
Any 5 of:	1–5
1. Start: first 10 seconds CrP or PCr	
2. After first 10-30seconds: glycogen or glucose	
3. After 30 seconds to 2 minutes fats are dominant fuel source	
4. Glycogen sparing will occur earlier in race (increased use of fats as fuel source)	
5 Consequence: Without supplementation of glucose his glycogen will deplete and therefore he must slow down to wait for energy from fat metabolism	
*May state increased protein metabolism without glycogen available	
	Total 5

Question 32

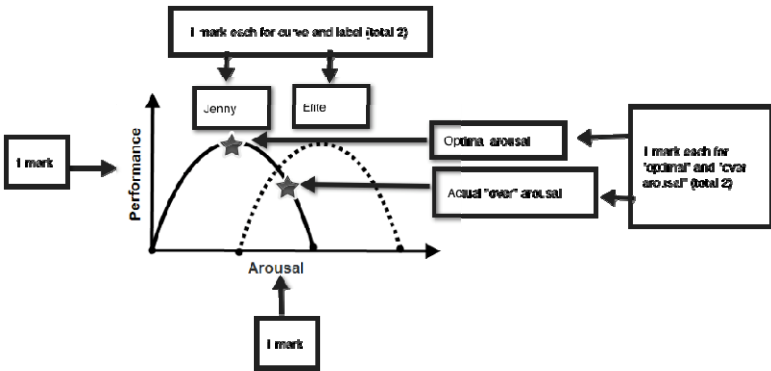
(15 marks)

Occasionally swimmers (such as Jenny in the image below) lose balance while on the starting block, causing them to make a false start.

- (a) Demonstrate your understanding of the biomechanical principle of balance by outlining **three** factors that influence stability while on the starting blocks and describe how each factor could be modified to improve Jenny's stability. (6 marks)

Description	Marks
Base of support	1
Description of <u>increasing</u> base of support by changing stance; e.g. track start	1
Line of gravity (or COM or COG)	1
Description of maintaining <u>line of gravity within base of support</u>	1
Height of COG or COM	1
Description of <u>lowering COG</u> to increase stability	1
Total 6	

- (b) Jenny is participating in her first senior competition. This is causing her anxiety during her performance. Demonstrate your knowledge of the inverted U hypothesis by drawing a labelled diagram below. Include an arousal curve for Jenny and a second arousal curve that represents an elite swimmer. Indicate where an ideal arousal level is for Jenny on her curve and where her actual arousal level was for the performance demonstrated by the image on page 23. (6 marks)

Description	Marks
 NB: wrong order 1 mark only	1–6
Total 6	

- (c) Describe how age, skill level and type of activity can influence arousal regulation. (3 marks)

Description	Marks
Age – e.g. younger athletes are easily over aroused (or similar/opposite statement)	1
Skill – e.g. more skilled athletes can handle higher arousal levels (or similar/opposite statement)	1
Type – activities that involved more complex skills require a lower level of arousal compared to activities with more simple skills	1
Total 3	

Question 33

(15 marks)

Novak Djokovic won the Australian Open Tennis championship in 2012. His success was not only a result of his skilful execution of his shots, but also the efficiency of his cardiorespiratory system.

- (a) Describe **three** immediate responses of Novak's respiratory system during a game of tennis. (3 marks)

Description	Marks
Any 3 of:	1–3
Increased ventilation (respiratory rate)	
Increased tidal volume	
Maximum oxygen uptake	
Increased gas exchange	
Increased a-VO ₂ difference (allowed in both circulatory and respiratory)	
	Total 3

- (b) From his time as a junior tennis player to his Australian Open win, describe **six** long-term adaptations of the circulatory system that would have occurred to Novak from his years of training. (6 marks)

Description	Marks
Any 6 of:	1–6
Increased cardiac output	
Decreased resting blood pressure	
Increased stroke volume (or $HR \times CO = SV$)	
Increased blood volume	
Increased capillaries	
Increased Hb content	
Increased a-VO ₂ difference (allowed in both circulatory and respiratory)	
Increased cardiac hypertrophy	
Decreased resting heart rate	
	Total 6

- (c) Name the **two** stages, and describe **two** characteristics of each stage, of the Fitts and Posner model of motor learning that Novak would have passed through successfully during his junior tennis years before he became a professional. (6 marks)

Description	Marks
Associative stage:	1
Refines the accuracy of performing the required skill (focus on correct timing/ tempo/ sequence) Reduces the degree of errors Begins to recognise errors Knows 'how to do' rather than 'what to do' Demonstrates performance is getting close to automation Requires frequent coaching feedback	1–2
Cognitive stage:	1
Needs to know what has to be done Needs to know how equipment is to be used Makes frequent large errors Forms a rough mental plan Requires a great deal of feedback Focus should be on individual skill rather than team skills Fatigue sets in quicker	1–2
	Total 6

ACKNOWLEDGEMENTS

Section Two

- Question 21** Abhyasika. (n.d.). *Gas exchange in the circulatory system* [Image]. Retrieved April 14, 2012, from www.abhyasika.com/10/10EM/10EMSandT/10EMSandT10/CapillaryNetwork.JPG.
- Question 24** Adapted image Shortlisted from: Stellar season sees Pearson leap into award contention. (2011, October 4). *Brisbane Times*. Retrieved April 14, 2012, from www.brisbanetimes.com.au/sport/. Copyright Getty Images.
- Question 26** Graph of three energy systems by courtesy of the examining panel.
- Question 27** Graph of 100 metre sprint performance by courtesy of the examining panel.
- Question 29** *Muscle anatomy* [Image]. (n.d.). Retrieved April 14, 2012, from www.shapesense.com/images/blank-muscle-anatomy.jpg.

Section Three

- Question 32(b)** Diagram of the inverted U hypothesis by courtesy of the examining panel.